
📘 Engineering Principles

C3 Contract Control Center

Engineering Principles

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Purpose

This document defines the engineering principles that govern development within the C3 Contract Control Center.

Unlike architecture documents, these principles describe *how* software should be developed rather than *how* it is structured.

All future development should be evaluated against these principles before implementation.

Principle 1

Architecture Before Features

New functionality shall not compromise architectural consistency.

If a feature requires bypassing established architectural boundaries, the architecture should be improved rather than circumvented.

Principle 2

Business Logic Exists Once

Business rules must have a single authoritative implementation.

Examples include:

- KPI calculations
- Renewal eligibility
- Workflow semantics
- Operational insights

Duplicate business logic is prohibited.

Principle 3

DTOs Are The Language Of The Application

The application communicates using domain models.

Infrastructure models shall never propagate beyond the Translation Layer.

Principle 4

Infrastructure Is Replaceable

No component should depend on SharePoint-specific behavior.

Infrastructure may evolve independently through adapters.

Principle 5

Presentation Is Passive

Screens present information.

They do not calculate business rules.

Business decisions belong within the domain layer.

Principle 6

Validate Before Expanding

Existing functionality should be validated before introducing additional features.

Every major domain should undergo relationship, navigation, KPI, and empty-state validation before platform expansion.

Principle 7

Documentation Is Part Of The Product

Architecture decisions, validation results, and significant design changes must be documented.

Documentation is maintained alongside the application rather than after development.

Principle 8

Optimize For Maintainability

Engineering decisions should prioritize long-term clarity over short-term implementation speed.

Code is expected to be read far more frequently than it is written.

Principle 9

Intelligence Over Reporting

The platform exists to support operational decision-making.

Analytics should evolve toward actionable guidance rather than passive dashboards.

Principle 10

Platform Before Module

Whenever new functionality is proposed, evaluate whether it belongs as:

- A reusable platform capability, or
- A domain-specific implementation.

Shared capabilities should be preferred whenever practical.

Summary

C3 is developed as an operational platform rather than a collection of pages.

These principles exist to ensure that future development preserves architectural consistency, maintains high engineering standards, and supports long-term platform evolution.
